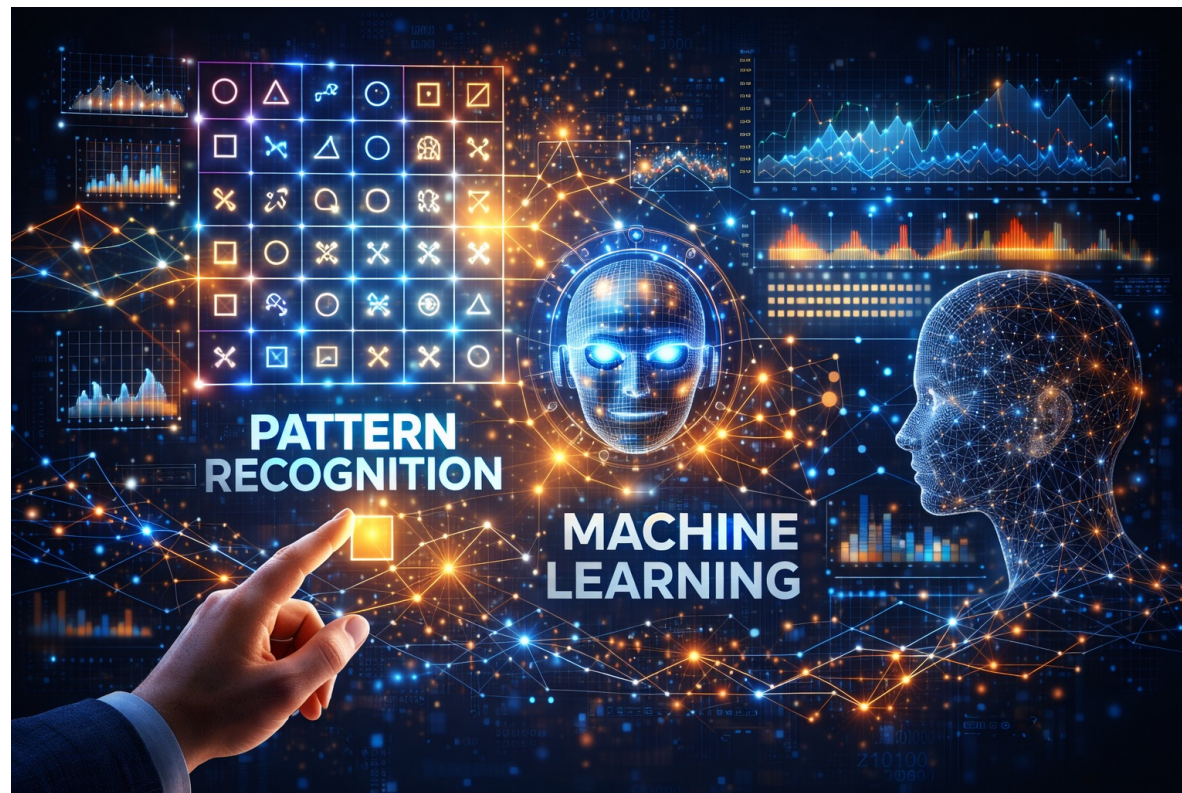


Introduction to Machine Learning

CSL2050 - Sem 02 – 2026

Lecture 16-17: Intro to Linear Discriminant Functions



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Lecture Plan

- Bayesian Decision Theory
- Minimum Risk Classification
- Gaussian Distribution and Mahalanobis Metric
- Discriminant Functions (DF's)
 - Multi-category
 - Two-category



Bayesian Decision Theory

- Bayes Formula

$$posterior = \frac{\textit{likelihood} \times \textit{prior}}{\textit{evidence}}$$

$$P(\omega_j|x) = \frac{p(x|\omega_j) P(\omega_j)}{p(x)} = \frac{p(x|\omega_j) P(\omega_j)}{\sum_{j=1,2} p(x|\omega_j) P(\omega_j)}$$

- **Bayes Decision Rule**

Decide ω_1 if $P(\omega_1|x) > P(\omega_2|x)$; Otherwise decide ω_2

Minimum-Error-Rate Classification

- Let true state of nature be ω_j and α_i be the possible action (of assigning data point $\mathbf{x}$ to specific ω_i) based up on observation $\mathbf{x}$.

- Let's define **Loss** associated with taking a specific action as:

$$\lambda_{ij} = \lambda(\alpha_i | \omega_j) = \begin{cases} 0 & i = j \\ 1 & i \neq j \end{cases}$$

- An expected loss is also called **Risk**. Conditional risk is defined as:

$$R(\alpha_i | \mathbf{x}) = \sum_{j=1}^c \lambda(\alpha_i | \omega_j) P(\omega_j | \mathbf{x})$$

Minimum-Error-Rate Classification

- Conditional Risk can be expressed as:

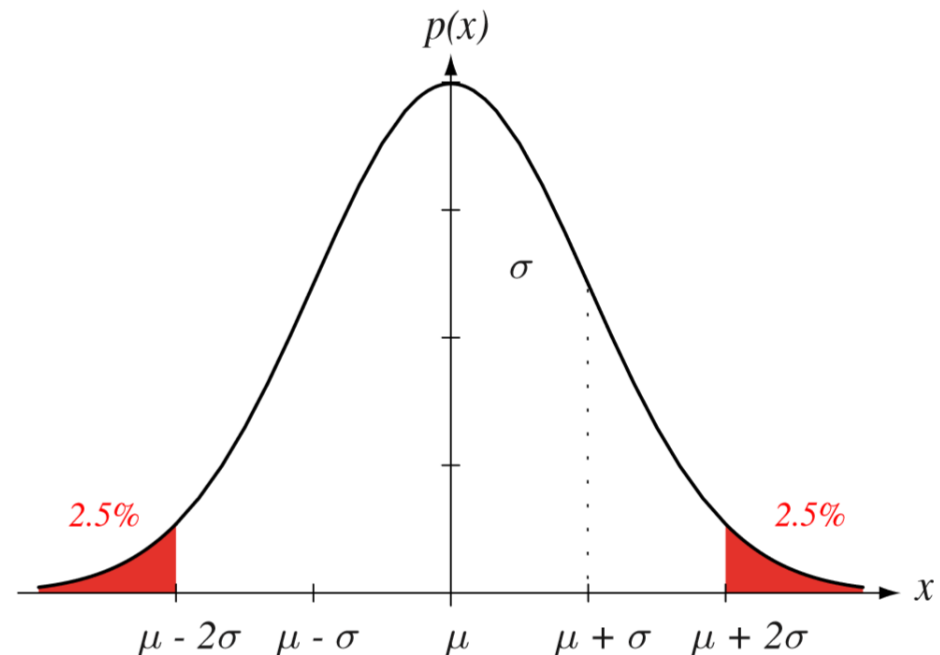
$$\begin{aligned} R(\alpha_i|\mathbf{x}) &= \sum_{j=1}^c \lambda_{ij} P(\omega_j|\mathbf{x}) \\ &= \sum_{j \neq i} P(\omega_j|\mathbf{x}) = 1 - P(\omega_i|\mathbf{x}) \end{aligned}$$

- If we choose ω_i corresponding to largest $P(\omega_i|\mathbf{x})$ then we minimize $R(\alpha_i|\mathbf{x})$
 - Decision rule:
Decide ω_i if $P(\omega_i|x) > P(\omega_j|x) \quad \forall j \neq i$
-

Univariate Normal Density

- Gaussian or normal density function is most popular due to:
 - Simple model with only two parameters
 - Central limit theorem
 - Closer to real world sampling of data
 - Makes less number of assumptions (maximum entropy)
 - Analytical tractability of mathematical form

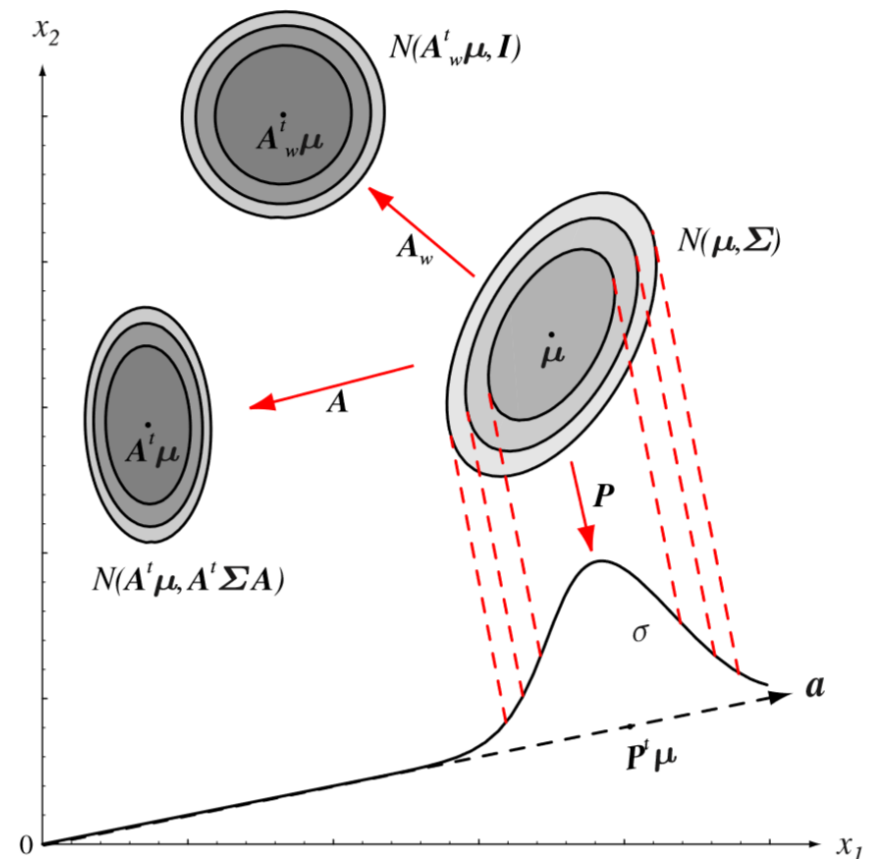
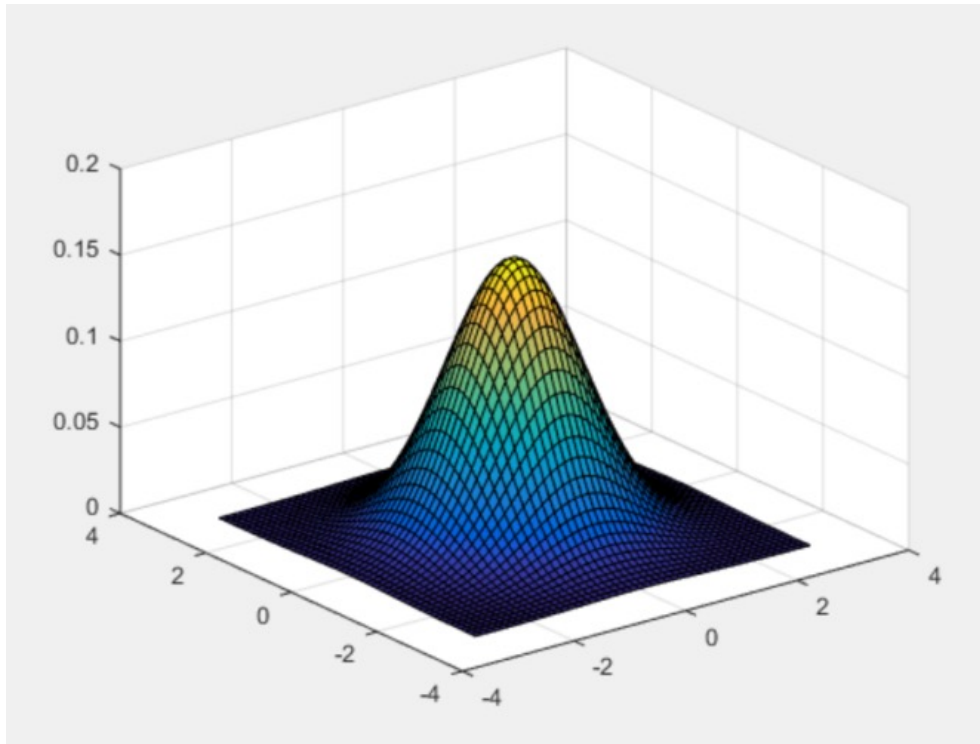
$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right] = \mathcal{N}(\mu, \sigma)$$



Multivariate Normal Density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right] = \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

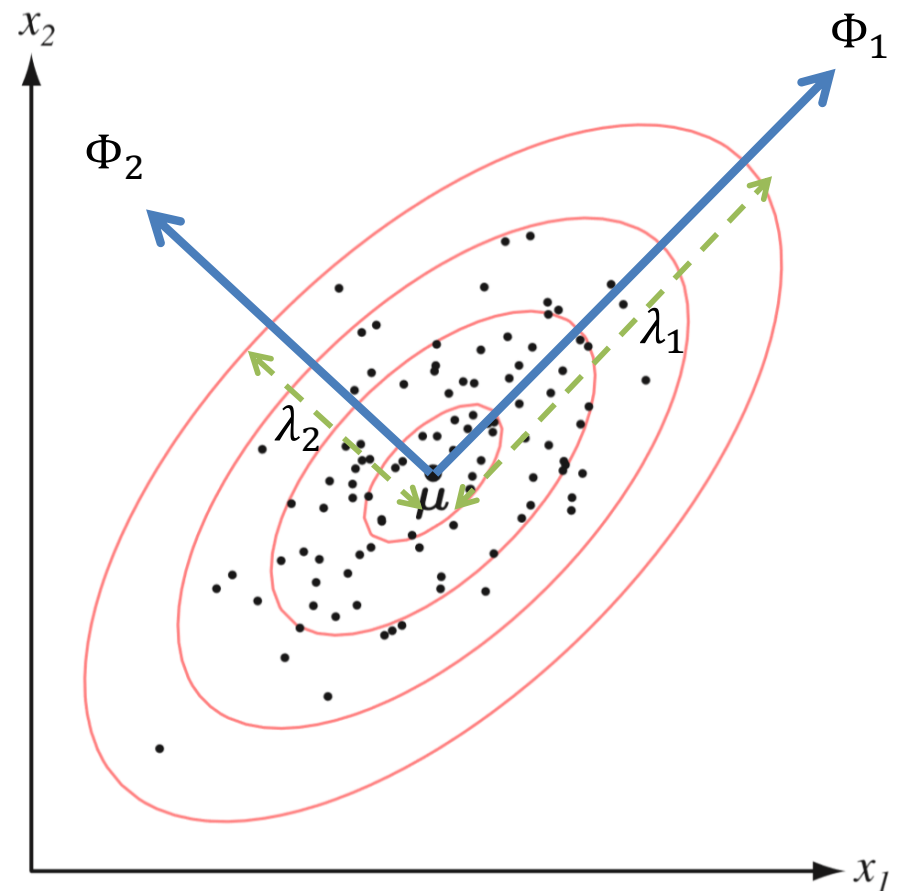
- Σ is a symmetric and positive semi-definite matrix so that $|\Sigma| > 0$.



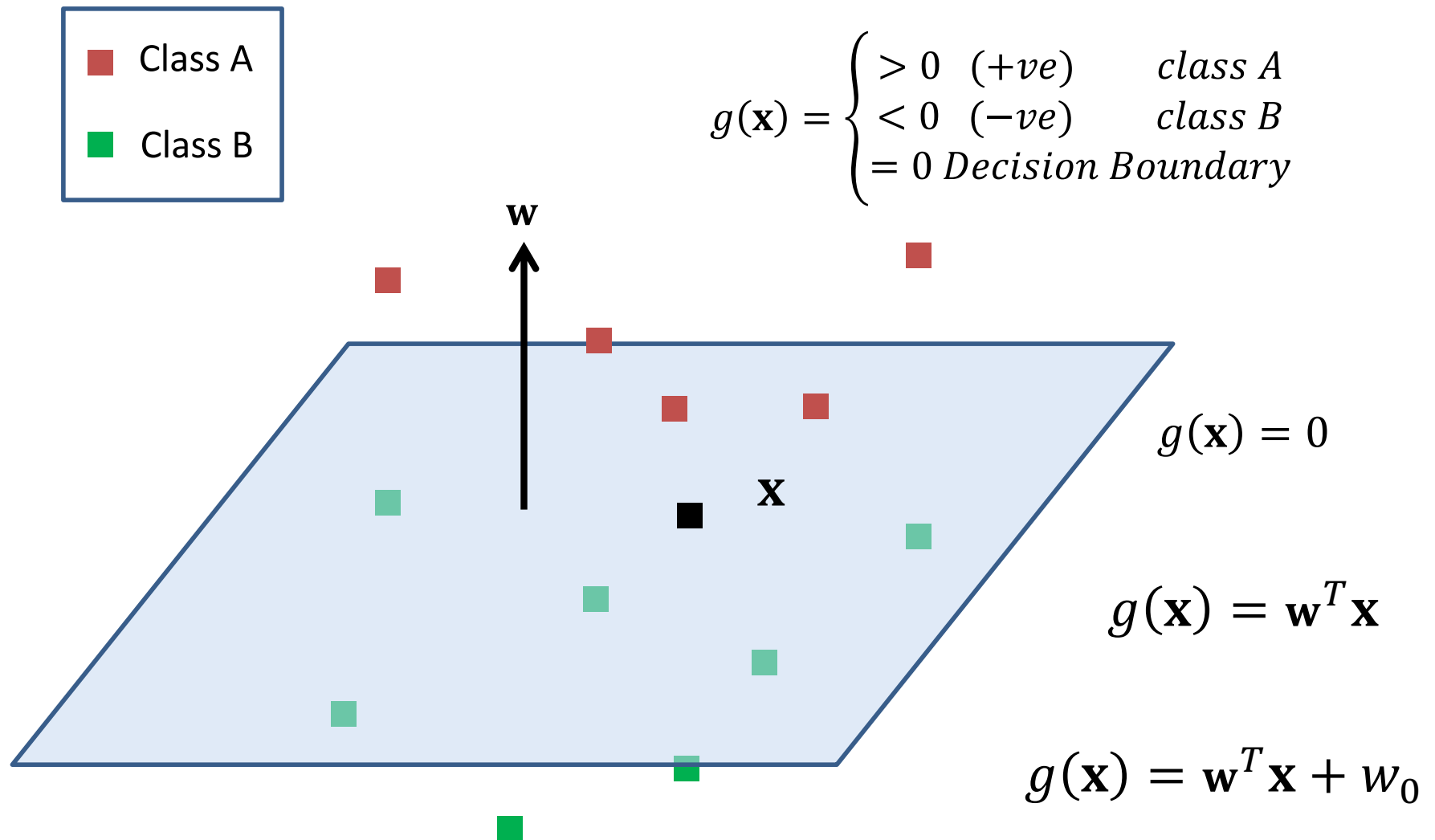
Mahalanobis Distance Metric

- $\Sigma = \Phi\Lambda\Phi^T$ where $\Phi = [\Phi_1, \dots, \Phi_d]$,
 $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$
- $r^2 = (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})$ is also known as Squared Mahalanobis Distance.
- $r^2 = (\mathbf{x} - \mathbf{y})^T S^{-1} (\mathbf{x} - \mathbf{y})$ is generalized form for computing distance between two RVs.
- Mahalanobis distance becomes Euclidean when $S = I$.

Prasanta Chandra
Mahalanobis

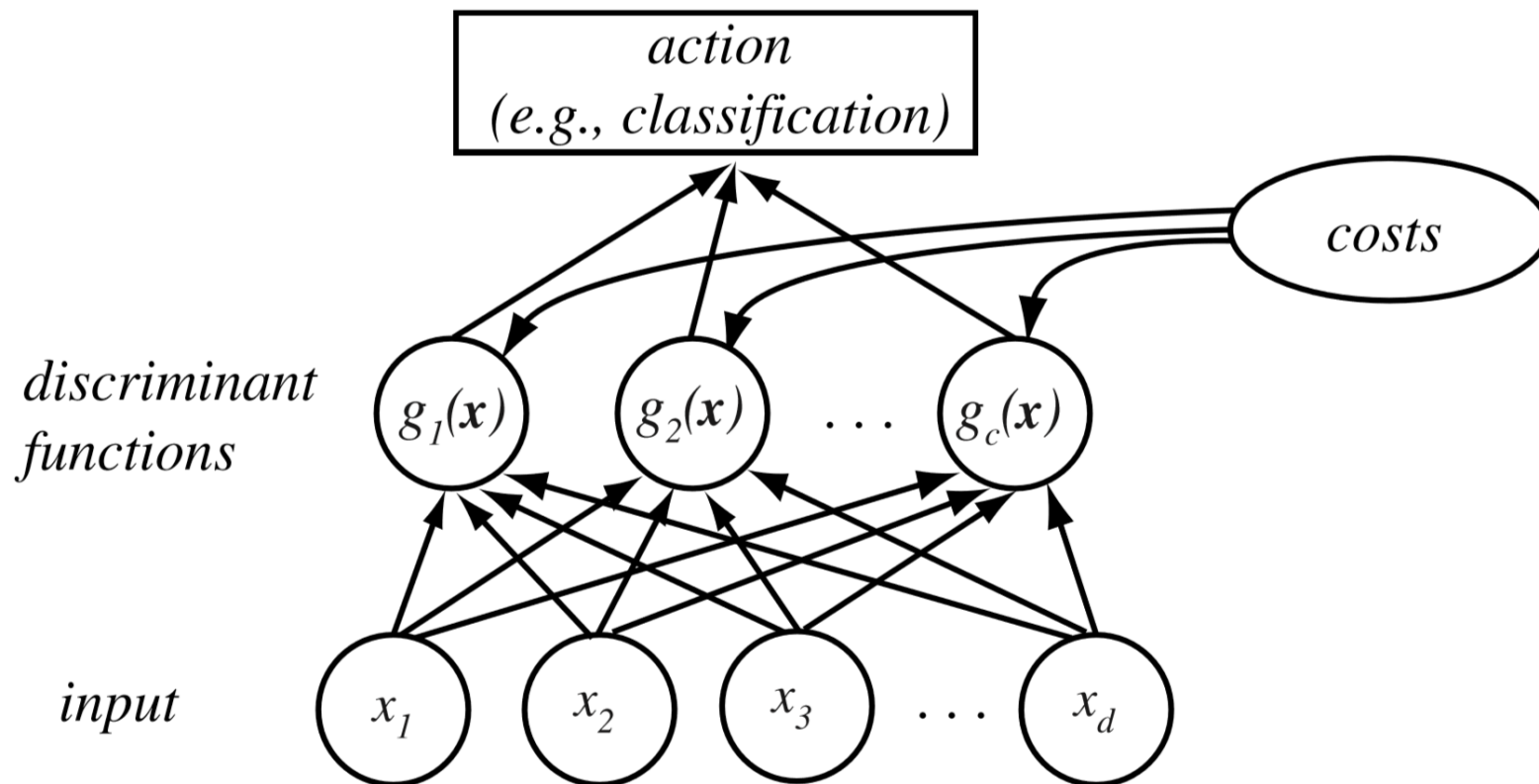


Linear Discriminant Functions & Decision Surfaces



Multi-category Discriminant Functions

- Assign class label ω_i to data point $\mathbf{x}$ if
$$g_i(\mathbf{x}) > g_j(\mathbf{x}) \quad \forall j \neq i \text{ and } i, j \in \{1, \dots, c\}$$



Multi-category Discriminant Functions

- Bayes Classifier as DFs: $g_i(\mathbf{x}) = -R(\alpha_i|\mathbf{x})$
- Or, $g_i(\mathbf{x}) = P(\omega_i|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_i) P(\omega_i)}{\sum_{j=1:c} p(\mathbf{x}|\omega_j) P(\omega_j)}$
- Or, $g_i(\mathbf{x}) = p(\mathbf{x}|\omega_i) P(\omega_i)$
- Let $f(\cdot)$ be a monotonically increasing function, e.g., $\ln$ (log with base e), then
$$f(g_i(\mathbf{x})) = \ln p(\mathbf{x}|\omega_i) + \ln P(\omega_i)$$
- Irrespective of mathematical form of DF's, the decision rules are equivalent.

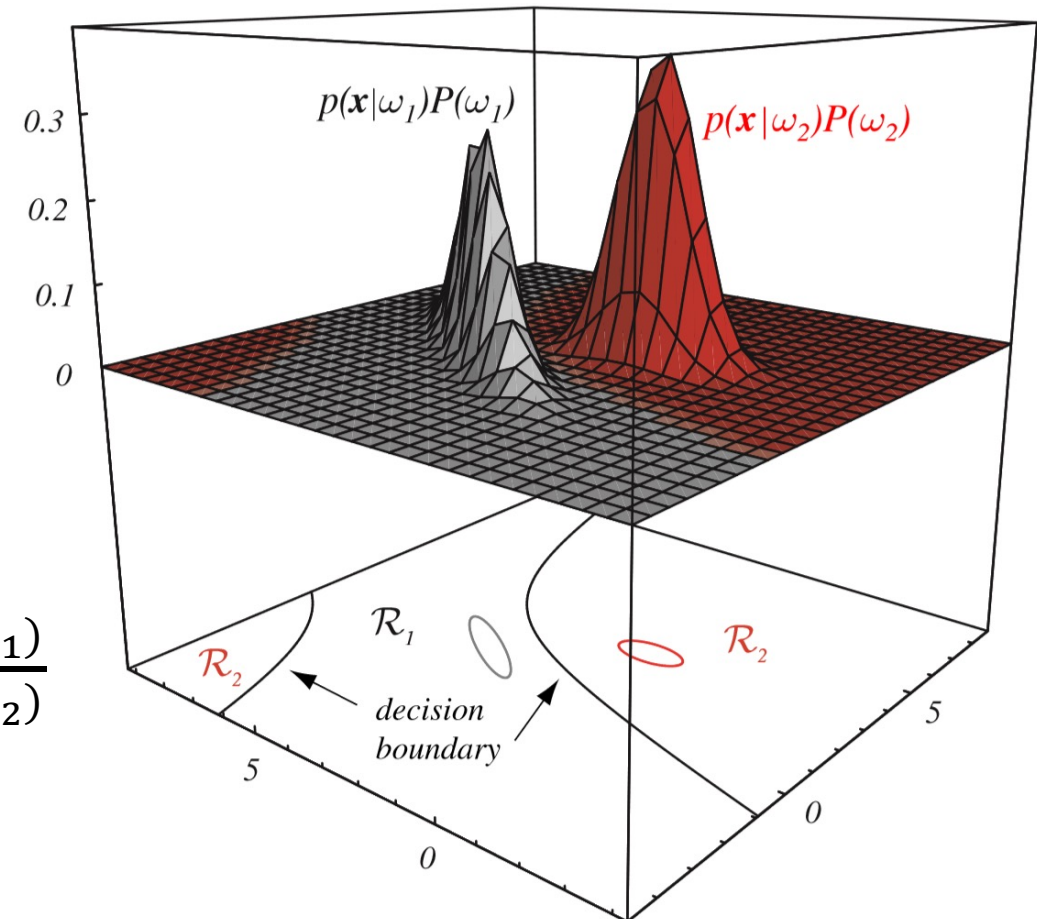
Two-category Discriminant Functions

- DICHOTOMIZER

Choose ω_1 if $g(\mathbf{x}) > 0$ where,
 $g(\mathbf{x}) = g_1(\mathbf{x}) - g_2(\mathbf{x})$

- $g(\mathbf{x}) = P(\omega_1|\mathbf{x}) - P(\omega_2|\mathbf{x})$

- $f(g(\mathbf{x})) = \ln \frac{p(\mathbf{x}|\omega_1)}{p(\mathbf{x}|\omega_2)} + \ln \frac{P(\omega_1)}{P(\omega_2)}$



Decision boundaries are two hyperbolas. Here ellipses mark where the density is $1/e$ times that at the peak of the distribution.

DF's for the Normal Density

- DF's: $g_i(\mathbf{x}) = \ln p(\mathbf{x}|\omega_i) + \ln P(\omega_i)$
- Let $p(\mathbf{x}|\omega_i)$ be Normal multivariate density, i.e., $p(\mathbf{x}|\omega_i) \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, then

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\boldsymbol{\Sigma}_i| + \ln P(\omega_i)$$

DF's for the Normal Density

- Case 1: Hyperspherical Clusters

- ❖ $\Sigma_i = \sigma^2 \mathbf{I}$

- $|\Sigma_i| = \sigma^{2d}$

- $\Sigma_i^{-1} = (1/\sigma^2)\mathbf{I}$

- ❖ $g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T (1/\sigma^2)\mathbf{I}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln \sigma^{2d} + \ln P(\omega_i)$

- ❖ $g_i(\mathbf{x}) = -\frac{\|\mathbf{x} - \boldsymbol{\mu}_i\|^2}{2\sigma^2} + \ln P(\omega_i)$

DF's for the Normal Density

- Linear DF:

$$g_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} + w_{i0}$$

where, $\mathbf{w}_i = \frac{1}{\sigma^2} \boldsymbol{\mu}_i$

and $w_{i0} = -\frac{1}{2\sigma^2} \boldsymbol{\mu}_i^T \boldsymbol{\mu}_i + \ln P(\omega_i)$

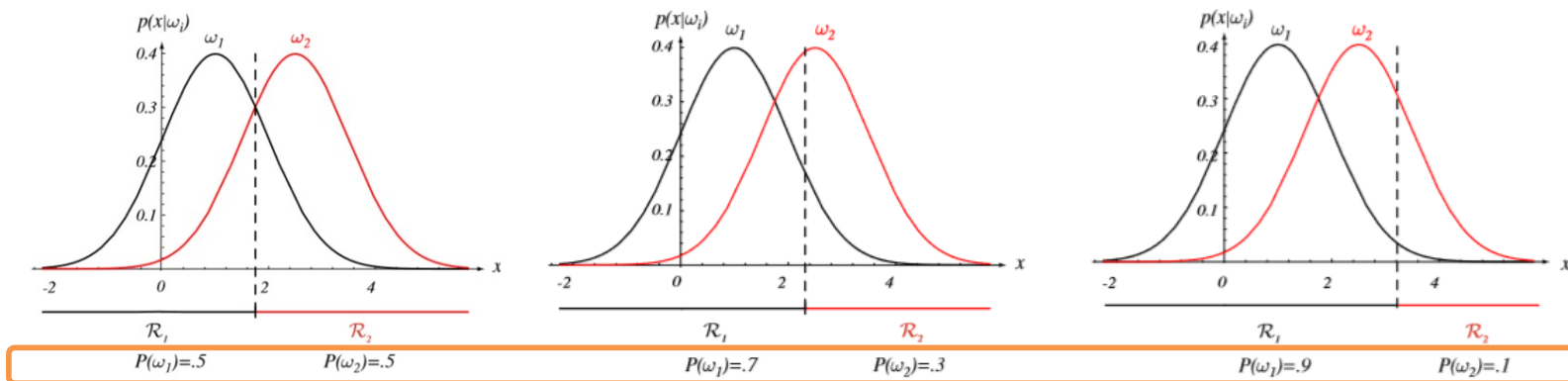
- Two-category case:

$$\mathbf{w}^T (\mathbf{x} - \mathbf{x}_0) = 0$$

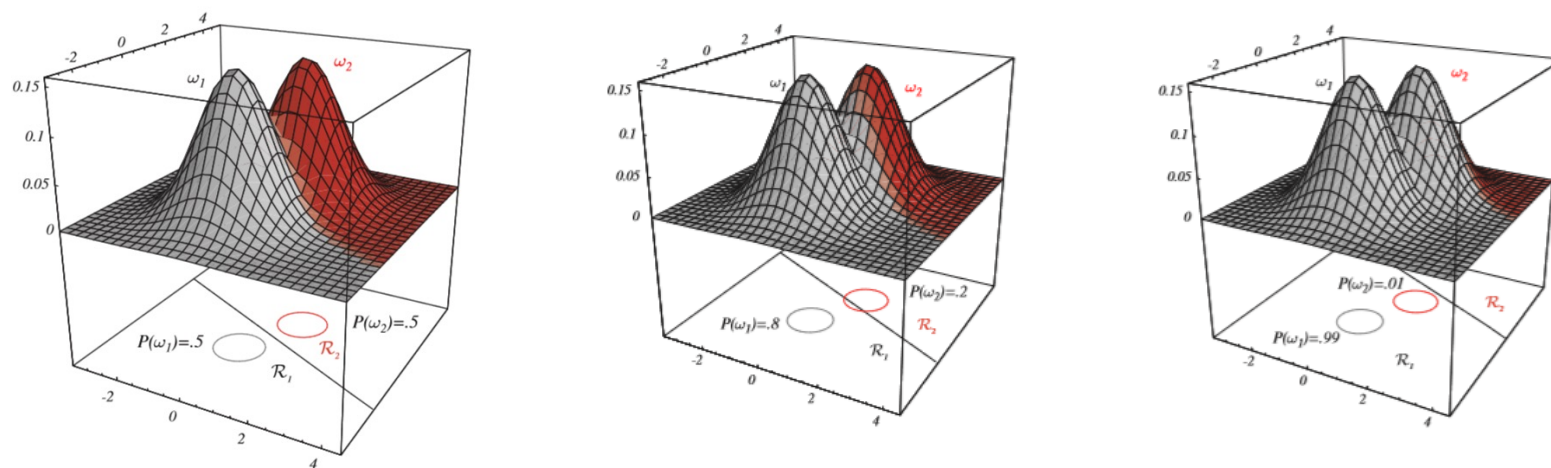
where, $\mathbf{w} = (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$

and $\mathbf{x}_0 = \frac{1}{2}(\boldsymbol{\mu}_i + \boldsymbol{\mu}_j) - \frac{\sigma^2}{\|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|^2} \ln \frac{P(\omega_i)}{P(\omega_j)} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$

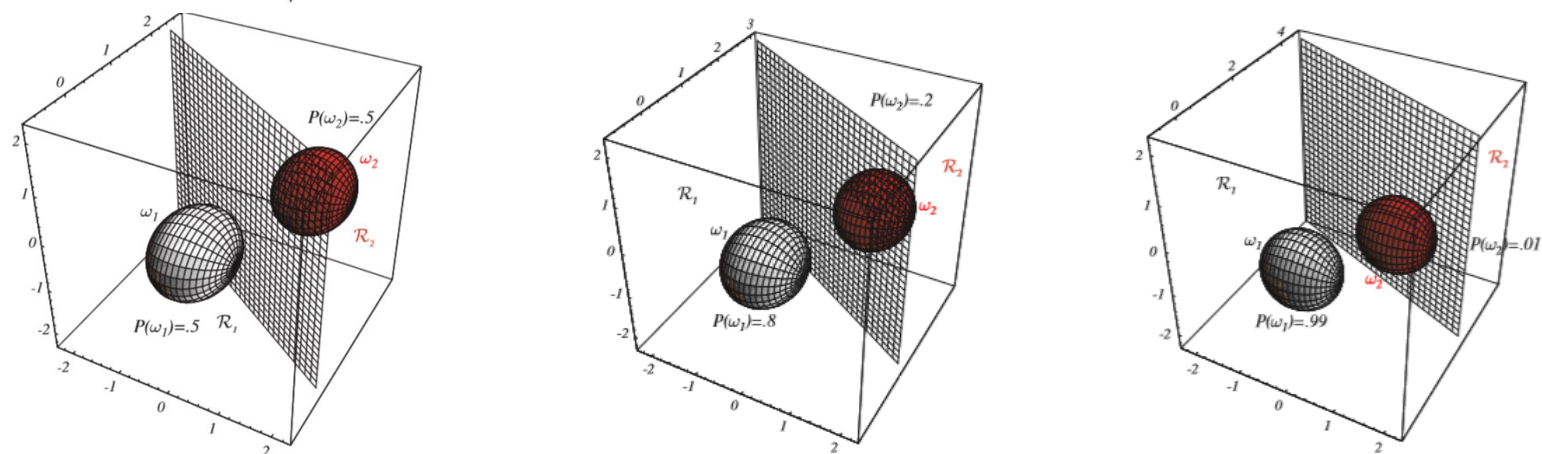
1D



2D



3D



DF's for the Normal Density

- Case 2: Hyperellipsoidal Clusters

$$\diamond \Sigma_i = \Sigma$$

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\boldsymbol{\Sigma}| + \ln P(\omega_i)$$

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) + \ln P(\omega_i)$$

DF's for the Normal Density

- Linear DF:

$$g_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} + w_{i0}$$

where, $\mathbf{w}_i = \Sigma^{-1} \boldsymbol{\mu}_i$

and $w_{i0} = -\frac{1}{2} \boldsymbol{\mu}_i^T \Sigma^{-1} \boldsymbol{\mu}_i + \ln P(\omega_i)$

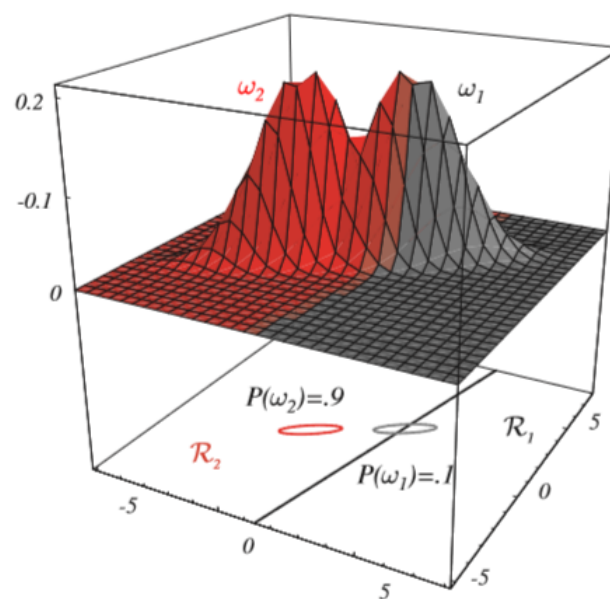
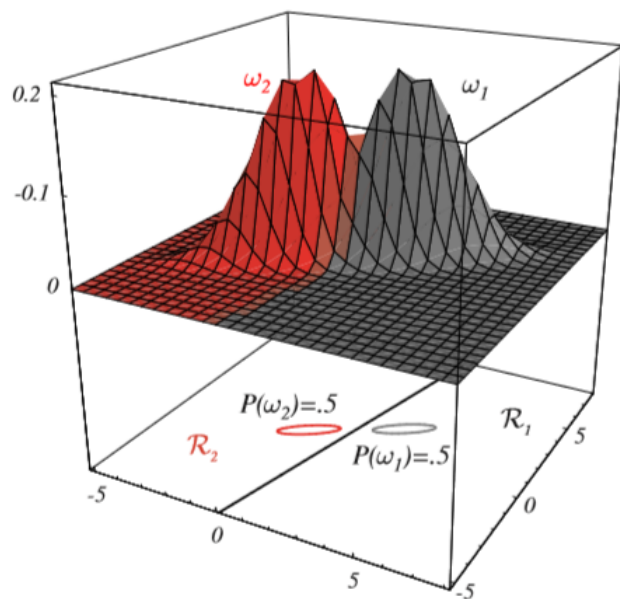
- Two-category case:

$$\mathbf{w}^T (\mathbf{x} - \mathbf{x}_0) = 0$$

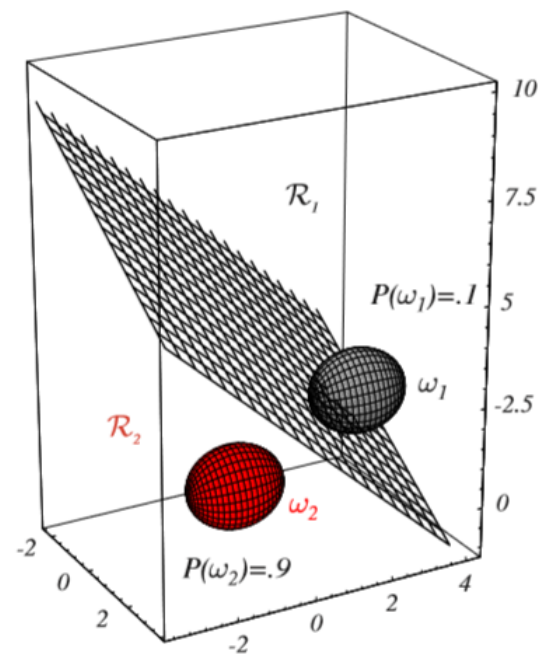
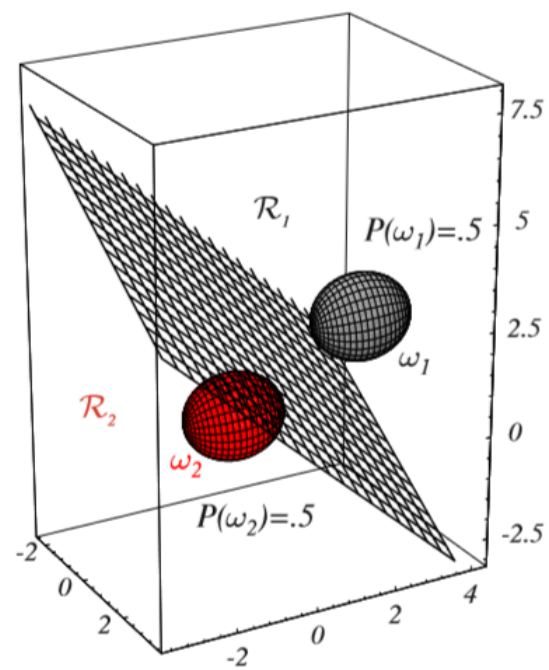
where, $\mathbf{w} = \Sigma^{-1} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$

and $\mathbf{x}_0 = \frac{1}{2} (\boldsymbol{\mu}_i + \boldsymbol{\mu}_j) - \frac{\ln [P(\omega_i)/P(\omega_j)]}{(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)^T \Sigma^{-1} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$

2D



3D



DF's for the Normal Density

- Case 3: Hyperquadrical Clusters

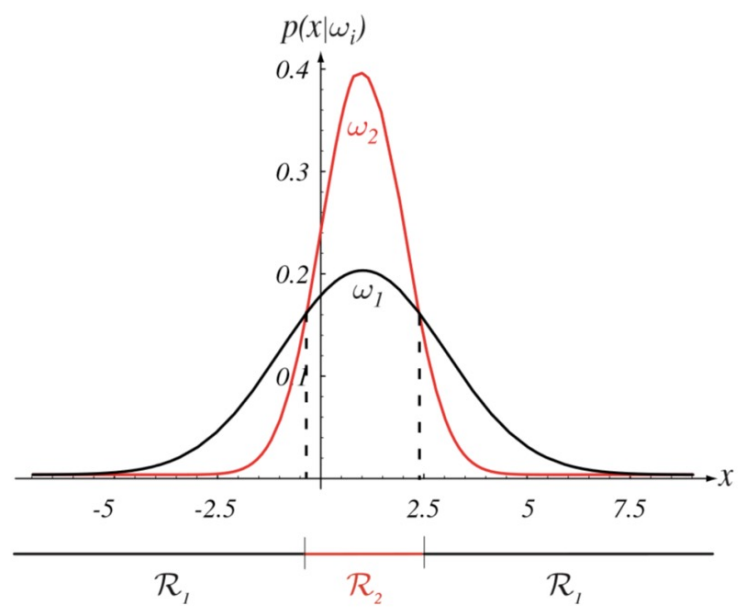
❖ Σ_i = arbitrary

- Linear DF

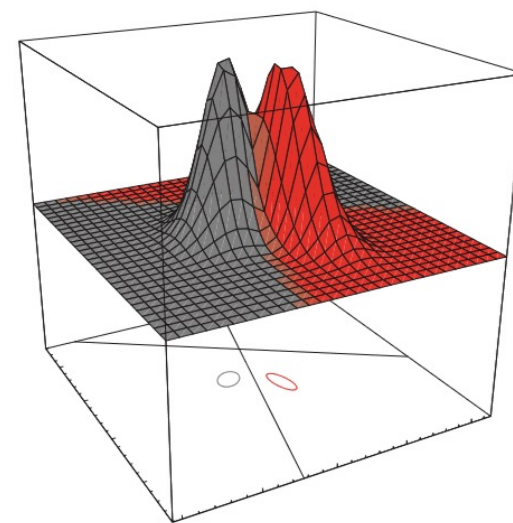
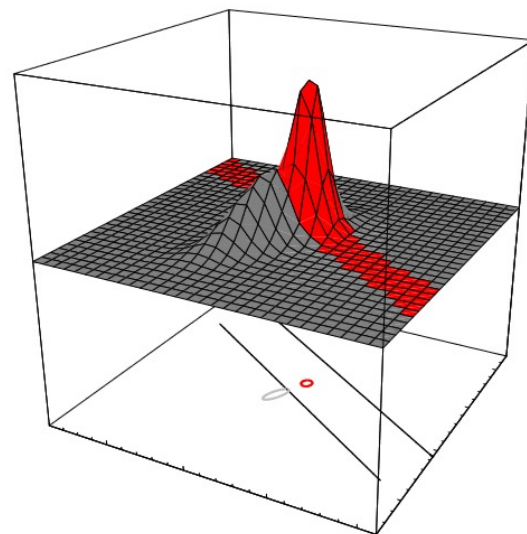
$$g_i(\mathbf{x}) = \mathbf{x}^T \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^T \mathbf{x} + w_{i0}$$

where, $\mathbf{W}_i = -\frac{1}{2}\Sigma_i^{-1}$, and $\mathbf{w}_i = \Sigma_i^{-1}\boldsymbol{\mu}_i$,

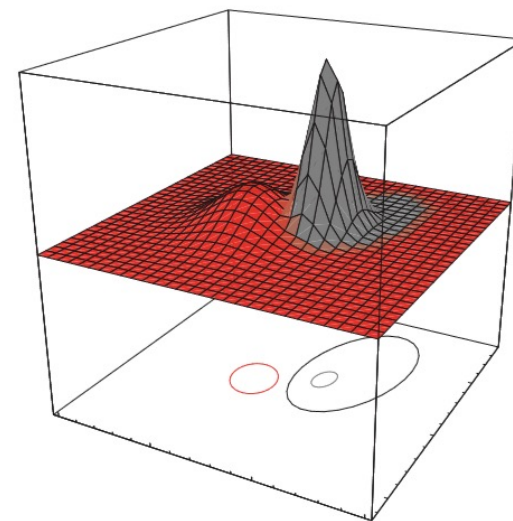
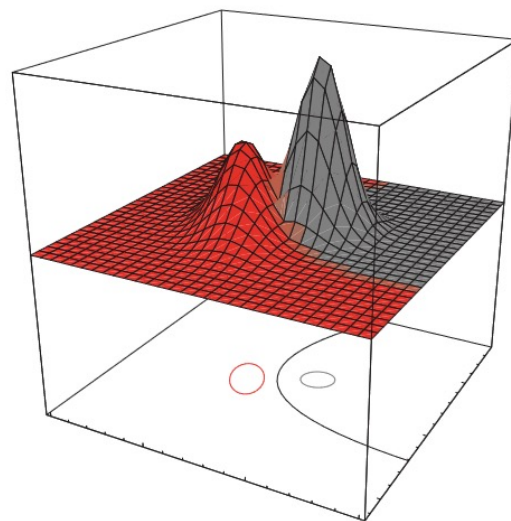
and $w_{i0} = -\frac{1}{2}\boldsymbol{\mu}_i^T \Sigma_i^{-1} \boldsymbol{\mu}_i - \frac{1}{2}\ln |\Sigma_i| + \ln P(\omega_i)$

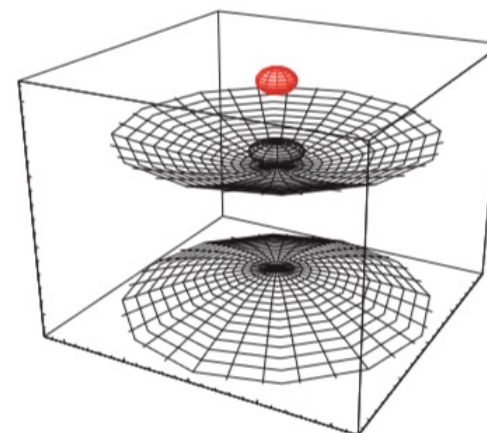
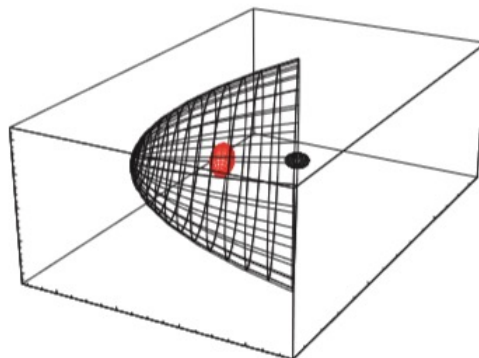
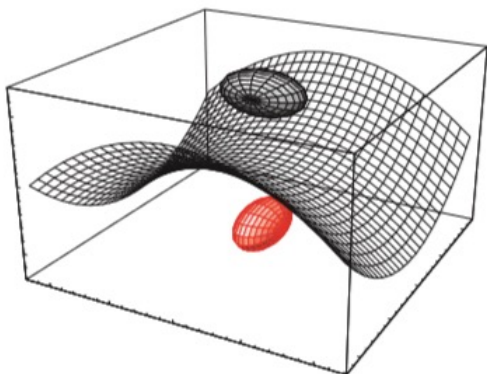
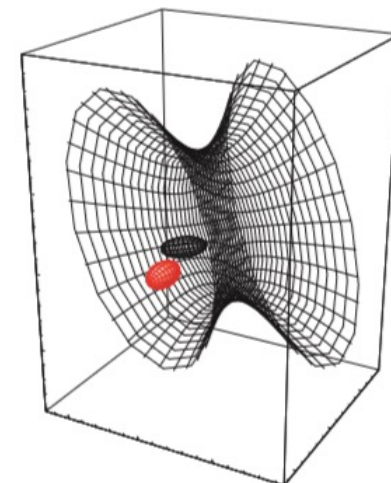
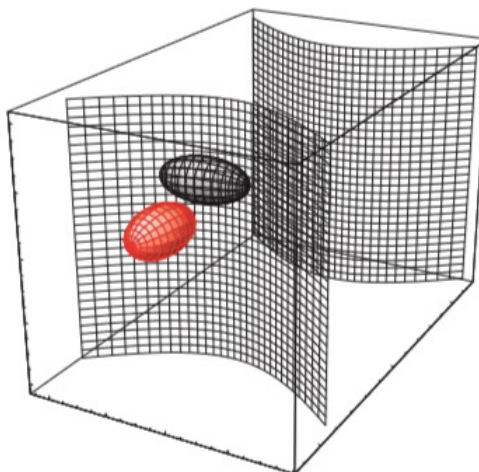
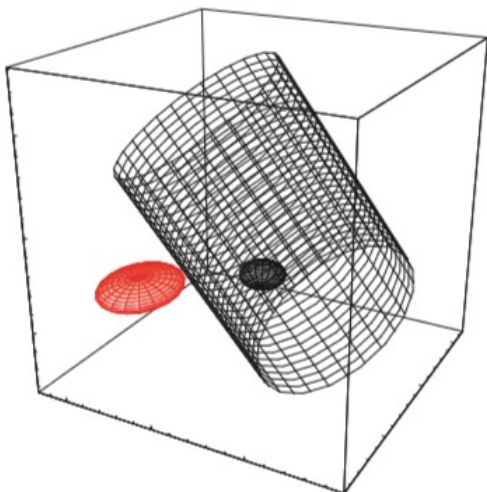
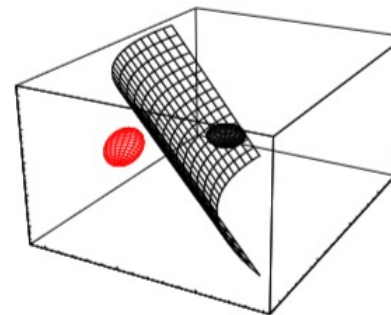
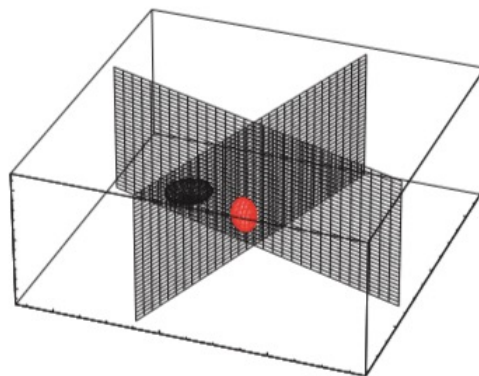
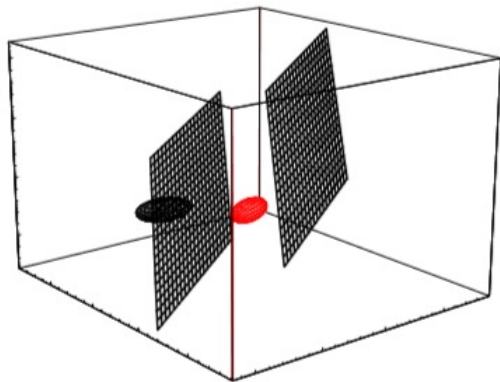


1D



2D





3D

Thank You !

